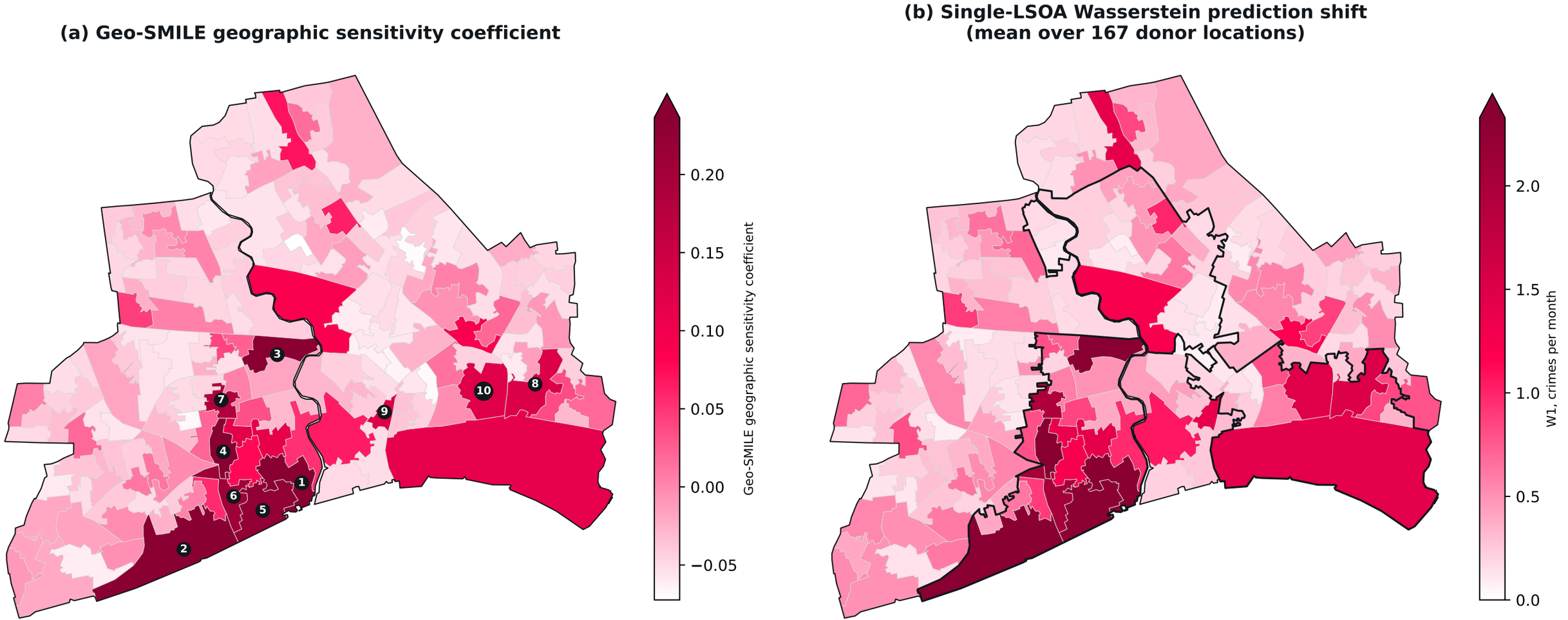


Random-donor design: XGBoost on 30 predictors + BNG easting/northing; masking an LSOA relocates its rows to the centroids of random other Hull LSOAs (K = 50 draws), all predictors kept. N = 4,000, seed 42. Top 10 numbered in (a); (b) black outlines = expert regions (map 1).
Sequential scales capped at the 98th percentile; the sign is not a direction of change. Not causal effects.